

- Hide menu
- Module 8 Introduction
- Lesson 1: Additive Models and Trees
- Lesson 2: Support Vector Machines
- Lesson 3: k-Means Clustering
- Lesson 4: Neural Networks
- Module 8 Summative Assessment
- Graded Assignment: Module 8 Summative Assessment
- Module 8 Summary

Your grade: 92.30%

Your score: 92.30% / Your highest score: 92.30%

Module 8 Summative Assessment

Next item →

Review Learning Objectives

Instructions

1. How is regression typically used in a feedforward neural network?

By using a sigmoid activation function in the output layer.

By having a single output neuron.

By categorizing continuous variables into discrete classes.

Using a softmax activation function in the first layer.

Correct

Due

Attempts

Correct: For regression, a single output neuron is common to predict continuous values.

Submitted

Nov 15, 1:32 AM ET

Try again

2. What is a common challenge associated with k-means clustering?

It cannot process datasets with more than 10,000 data points.

It can be sensitive to the initial placement of the centroids.

It guarantees the same result in every run with the same data.

It always requires labeled data for clustering.

Correct

0 Use

Make

Report an issue

Correct: The initial placement of centroids can affect the final clustering solution.

3. What is the primary benefit of using an additive model over a more complex model?

It offers interpretability while still being flexible enough to model nonlinear relationships.

It always outperforms more complex models in terms of prediction accuracy.

It requires significantly less data to train effectively.

It automatically performs feature selection during model training.

Correct

Correct: Additive models are interpretable and can model nonlinear relationships in a flexible way.

4. What role does the parameter C play in a Support Vector Classifier?

It sets the threshold for removing outliers from the dataset.

It determines the trade-off between maximizing the margin and minimizing the classification error.

It specifies the number of support vectors to be used in the model.

It controls the degree of the polynomial kernel used in the SVM.

Correct

Correct: The C parameter balances the margin size with the misclassification penalty.

5. In the context of SVMs, what is a "margin"?

The minimum distance required to correctly classify all data points.

The distance between the separating hyperplane and the nearest data point from either class.

The probability that a given data point is correctly classified.

The angle between the support vectors and the hyperplane.

Correct

Correct: The margin is the gap between the hyperplane and the nearest points (support vectors) of each class.

6. What is the significance of choosing different kernel functions in SVM?

Kernel choice determines the maximum number of classes that can be separated by the SVM.

The kernel function changes the basic principle of how SVM operates.

Different kernels allow SVM to model complex and nonlinear boundaries between classes.

Different kernels significantly alter the computational complexity of the algorithm.

Correct

Correct: Kernel choice impacts the SVM's ability to handle nonlinear relationships.

7. Which of the following is a disadvantage of using tree models?

Tree models cannot handle categorical variables.

Tree models cannot be used for regression tasks.

They are less interpretable compared to linear regression models.

They can easily overfit the training data if not properly pruned or regularized.

Correct

Correct: Tree models can overfit if they grow too complex without constraints.

8. How does the kernel trick in SVM enhance its performance?

It reduces the dimensionality of the feature space to make computations faster.

It linearly separates the data points regardless of the original feature space.

It allows SVM to find a separating hyperplane in a higher-dimensional space without explicitly computing the transformation.

It directly increases the speed of the SVM algorithm.

Correct

Correct: The kernel trick facilitates the handling of nonlinear relationships by working in a higher-dimensional feature space.

9. What happens in the assignment step of the k-means algorithm?

The algorithm calculates the total within-cluster variation.

Each data point is assigned to the nearest cluster centroid.

The value of K is determined.

The centroids are moved to the center of their clusters.

Correct

Correct: In the assignment step, data points are assigned to the cluster with the closest centroid.

10. How does k-means clustering algorithm typically initialize the cluster centroids?

Usually by randomly selecting 'k' data points as the initial centroids.

By placing all centroids at the center of the dataset.

By dividing the dataset evenly into 'k' segments and using their centers.

By using the mean of the entire dataset as the starting point for all centroids.

Correct

Correct: A common approach is to randomly choose k data points as the initial centroids.

11. How do additive models handle interactions between variables?

They require manual specification of all possible interactions before modeling.

They automatically detect and model all interactions between variables.

Additive models cannot handle interactions and are limited to independent effects only.

Additive models generally assume no interactions, but interactions can be included if specified.

Incorrect

Incorrect: Additive models do not automatically include interactions.

12. What does the term "feedforward" imply about the architecture of a neural network?

The neurons in each layer are interconnected.

The input data is forwarded through a series of feedback loops.

The network feeds forward error information from the output back to the input layer.

The information flows from the input layer to the output layer without any backward or lateral connections.

Correct

Correct: "Feedforward" implies a unidirectional flow of data from input to output layers.

13. In a feedforward neural network for classification, what is typically the final layer's activation function?

ReLU in all types of classification tasks.

Softmax for multi-class classification or sigmoid for binary classification.

Linear activation function.

Hyperbolic tangent in all scenarios.

Correct

Correct: Softmax is used for multi-class, and sigmoid is typically used for binary classification tasks.